

Scale-Out vs. Scale-Up: A Design Space Exploration of Weight-Resident SIMD Neural Accelerators Under Memory Constraints

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Abstract—Edge-deployed neural network accelerators experience performance degradation when arithmetic array expansion outpaces available memory bandwidth. To quantify this limitation, this research executes a mixed-level design space exploration of a temporally folded, weight-resident Matrix-Vector Unit. The register-transfer level datapath undergoes 14 nanometer logic synthesis across 116 geometric configurations to extract operational frequency, silicon area, and power dissipation metrics. Applying these physical parameters to an Extended Roofline Model and Energy-Delay Product assessment evaluates system scalability under variable memory transfer rates. The extracted data indicates that expanding the vector datapath induces interconnect congestion, which lowers operational frequency. Conversely, increasing output channel depth replicates local storage logic, elevating static leakage power. Although high operational intensity insulates monolithic architectures from both bandwidth and latency limitations, their silicon area and power costs preclude embedded deployment. Under a 25 GB/s memory constraint, a distributed array of dimensionally restricted processing nodes achieves 104 GMAC/s, reducing power consumption by 82 percent and silicon area by 79 percent compared to a contiguous array operating at 32 GMAC/s. These findings establish that scale-out topologies exhibit superior energy-delay profiles relative to scale-up configurations, dictating the restriction of localized register capacities to optimize edge accelerator efficiency.

Index Terms—Matrix-Vector Unit, Extended Roofline Model, Energy-Delay Product, Spatial Dataflow

I. INTRODUCTION

Neural Network (NN) inference relies on multi-dimensional matrix operations, executed primarily as Multiply-Accumulate (MAC) functions [1]. Processing these extensive parameter sets necessitates continuous data retrieval, which imposes two distinct hardware limitations. First, transferring operands from off-chip DRAM requires orders of magnitude more energy than executing the operations themselves [2]. Furthermore, even moving data from internal on-chip caches consumes significantly more energy than the arithmetic computations, creating a severe power bottleneck [2]. Second, limited memory

bandwidth physically restricts computational throughput; even if an accelerator has the thermal budget to compute faster, arithmetic arrays sit idle awaiting data delivery [1]. Ultimately, empirical data from large-scale deployments confirms that the vast majority of neural network workloads remain fundamentally bottlenecked by memory bandwidth rather than computational capacity [3].

Cloud infrastructures resolve external data transfer limitations by integrating High-Bandwidth Memory modules, which leverage 3D stacking and silicon interposers to deliver massive bandwidth at a lower energy cost per bit [4]. However, the prohibitive packaging costs and complex physical integration requirements of such advanced memory architectures [4] preclude their adoption in edge computing environments, which operate under limited power and thermal envelopes. Restricted to conventional memory interfaces, embedded architectures incur operand delivery delays when executing high-dimensional models. Expanding local arithmetic logic units without a proportional increase in external memory bandwidth yields diminishing returns, as processing elements remain idle awaiting data transmission [1].

Optimizing hardware within these physical boundaries requires systematic Design Space Exploration. Isolating specific computational blocks enables engineers to quantify the interaction between logical scaling and memory bandwidth. This work investigates a parameterizable weight-resident Matrix-Vector Unit (MVU) implementing a temporally folded Single Instruction, Multiple Data (SIMD) datapath, derived from the FINN streaming architecture [5], [6]. By synthesizing distinct SIMD lane counts and output channel capacities across 116 configurations, this research provides a quantitative analysis of structural scaling limits. The primary contribution comprises empirical guidelines for dimensioning edge-oriented hardware, and a reusable analytical methodology for identifying the microarchitectural thresholds where computational expansion ceases to generate proportional throughput gains. Unlike traditional analytical DSEs, we synthesize 116 configurations using an academic 14nm FinFET process, grounding our performance, area, and power metrics in industry-standard synthesis workflows rather than high-level theoretical proxies.

This research was developed within the Computer Engineering program at the Universidade de São Paulo (USP). The authors gratefully acknowledge the Universidade Federal do Rio Grande do Sul (UFRGS) CADMicro facility for providing the server infrastructure, Electronic Design Automation (EDA) tools, and the PDK utilized in this work.

II. BACKGROUND AND RELATED WORK

A. Edge DNN Accelerators and Weight Residency

Spatial architectures such as Eyeriss and its v2 extension [7], [8] distribute arithmetic across 2D processing arrays with hierarchical routing to maximize operand reuse. Streaming architectures, by contrast, synthesize structurally independent hardware engines per network layer [9]. To sustain continuous execution, these streaming architectures rely on a permanent parameter residency strategy, an approach similarly deployed at cloud scale in Microsoft Project Brainwave [10]. Weight-resident architectures permanently allocate filter parameters to distributed on-chip storage. This strategy eliminates off-chip memory reads during inference, restricting external transfers to input activations [6], [10]. Utilizing this strategy, the FINN framework translates convolutional operations into layer-level matrix-vector products mapped onto parameterizable streaming engines called Matrix-Vector Units (MVUs) [5], [9]. When logical tensor dimensions exceed the physical capacity of the instantiated SIMD lanes, hardware logic time-multiplexes execution, accumulating partial sums across consecutive clock cycles to synthesize the complete matrix-vector product [5], [9]. Adjusting the degree of temporal folding permits engineers to balance parallel processing capabilities against silicon area constraints.

B. Analytical Models: Roofline and EDP

The Energy-Delay Product (EDP) quantifies computational efficiency by multiplying total energy dissipation by execution time [11]. This calculation penalizes configurations that trade power for throughput through frequency or voltage scaling [11]. While EDP assesses internal datapath execution, it omits macroscopic data transfer limitations. The Extended Roofline framework addresses this by graphing maximum achievable performance against the operational intensity of a workload [12], [13], defining the boundary where hardware transitions from memory-bound to compute-bound operation. Together, EDP isolates localized functional efficiency while the Extended Roofline identifies whether the external memory interface sustains the corresponding arithmetic output.

III. ARCHITECTURE AND METHODOLOGY

A. System-Level Architecture

Deep neural networks map onto cascaded MVUs, bypassing the conventional paradigm of writing intermediate activations to off-chip DRAM [5], [9]. Each unit computes a specific layer—or distributes output channels of the same layer across parallel nodes—and transmits resulting activations directly to the subsequent stage, as illustrated in Fig. 1. Aggregate pipeline throughput is, thus, bounded by the minimum execution rate among all nodes [5]. Insufficient operand delivery at any interface forces the corresponding arithmetic logic into idle states, motivating the characterization of bandwidth thresholds for an isolated MVU.

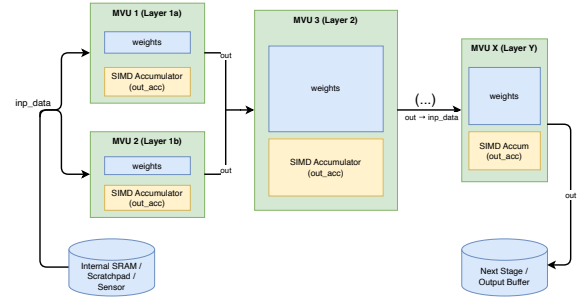


Fig. 1. Macroscopic view of a Deep Neural Network of Y layers running on X MVUs.

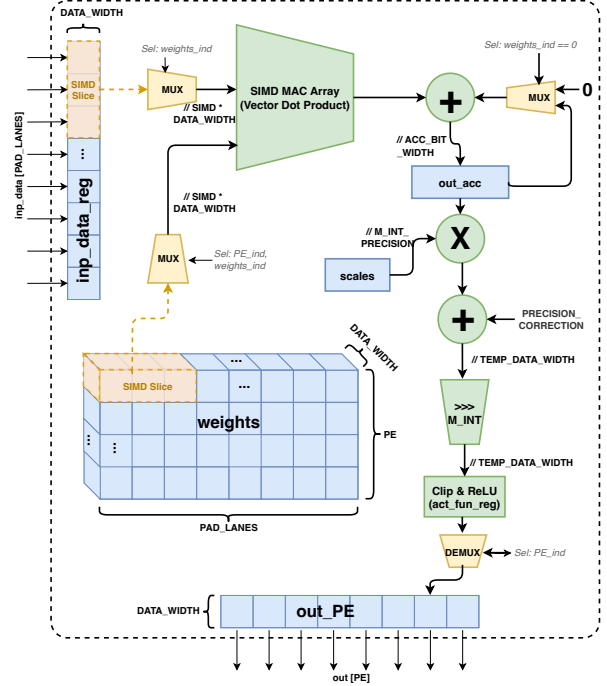


Fig. 2. Internal Architectural View of an MVU.

B. MVU Microarchitecture

The evaluated MVU, detailed in Fig. 2, is a parameterized compute engine governed by a Finite State Machine (FSM). Functional correctness of the bias-free, 8-bit integer quantized arithmetic was explicitly verified against a PyTorch-based golden model. Hardware computational structure is established through two primary parameters: the concurrent arithmetic vector width (SIMD) and the number of sequential output channel processing elements (PE) [6]. During initialization, the FSM populates internal register arrays with layer-specific filter parameters in SIMD-aligned segments, appending one fixed-point requantization scale. During inference, filter parameters remain static, restricting external memory requests exclusively to incoming feature maps. The computation phase multiplies each activation segment by the resident weights, accumulates partial sums, then applies fixed-point requantization and non-linear functions [5], [6].

C. Design Space and Synthesis Setup

Physical characterization relies on an automated logic synthesis pipeline. The RTL implementation is a simplified, single-datapath MVU core developed within the research group, conceived in accordance with the architectural model described in [6], and compiled against the Synopsys SAED 14nm FinFET standard cell library [14] using Design Compiler Graphical. Static Timing Analysis and power estimation execute via PrimeTime under nominal Process, Voltage, and Temperature conditions. All configurations were synthesized under a uniform 70 MHz clock target, optimizing for area and power; the reported $F_{\max}$ values therefore reflect post-synthesis slack recovery. This target frequency prevents per-configuration frequency optimization that would trigger excessive power and area inflation. Thus, $F_{\max}$ values provide a controlled comparative baseline rather than absolute theoretical peaks. Power is estimated under static switching activity conditions (`toggle_rate` = 0.1, `static_probability` = 0.5), which approximates typical toggle behavior but does not capture data-dependent transitions.

The design space exploration swept `SIMD` $\in \{4, 8, 16, 32, 64, 128, 256, 512, 1024\}$ and `PE` $\in \{8, 16, 32, 64\}$; a third variable, the maximum input tensor dimension, was co-varied per configuration to satisfy SIMD alignment, yielding 116 valid configurations. For each permutation, the toolchain extracts $F_{\max}$, integrated silicon area, and total power dissipation. Memory bandwidth constraints are modeled analytically: execution time is computed as $T = L + s/\text{BW}$, where L is the memory access latency, s is the data transfer size, and BW is the available bandwidth. This linear transfer equation assumes an ideal, SRAM-like unbuffered pipeline, intentionally excluding DRAM bank conflicts and intermediate FIFOs to isolate architectural bottlenecks. Because weights reside on-chip, Operational Intensity (OI) is defined as total MAC operations per byte of input activation read, scaling directly with the `PE` parameter (analog to what is done in [3]). Pipeline stalls are injected proportionally when memory transfer time exceeds datapath execution time, across three bandwidth profiles spanning the practical range of embedded memory interfaces — from severely constrained systems (1.0 GB/s, 20 ns) through mid-range edge SoCs (6.0 GB/s, 10 ns) to high bandwidth embedded platforms (25.0 GB/s, 1 ns) — chosen to expose the crossover point at which scale-out topologies overtake monolithic configurations. These idle cycle counts parameterize the Extended Roofline [13] boundaries, and the derived physical metrics feed the EDP formulation.

IV. RESULTS AND DISCUSSION

A. Synthesis Metrics

Logic synthesis reveals that expanding SIMD initially increases theoretical throughput, but at critical density thresholds induces interconnect congestion. This routing complexity extends the critical path, degrading $F_{\max}$ and negating the computational advantages of wider multiplier arrays.

Incrementing PE linearly replicates the localized weight register arrays, with non-combinational storage logic governing spatial scaling. In the densest configuration (PE=64, SIMD=512), parameter storage occupies 72.1% of total silicon area, rendering the arithmetic logic physically marginal. Static leakage consequently dominates the power profile of high-dimensional configurations. Normalizing throughput against resource utilization identifies the optimal geometry at SIMD=128 and PE=8, which sustains 254.45 MHz to deliver 26.06 GMAC/s while dissipating 82.60 μW , establishing the efficiency baseline.

To contextualize the subsequent system-level analysis, three representative configurations are extracted from these synthesis boundaries: a resource-constrained node (Model 1: PE=8, SIMD=32), the optimally balanced node (Model 2: PE=8, SIMD=128), and a monolithic scale-up node (Model 3: PE=64, SIMD=512).

B. Roofline Analysis

Projecting physical limits onto the Extended Roofline model exposes system-level scaling boundaries. Under a 25.0 GB/s memory interface, Model 3 is completely compute-bound: routing congestion caps $F_{\max}$ at 64.23 MHz, limiting throughput to 31.89 GMAC/s. Model 2 requires approximately 6.0 GB/s to saturate its arithmetic logic at 26.06 GMAC/s, as seen in Fig. 3. Distributing the available 25.0 GB/s bandwidth across four independent Model 2 nodes—an analytical projection reserving 1.0 GB/s as margin for unmodeled inter-node overhead—delivers 104.24 GMAC/s in a scale-out topology, superseding the monolithic array’s throughput by more than 3 times while reducing net silicon area by 79% and power consumption by 82%.

The monolithic paradigm retains an advantage exclusively under restricted bandwidth. At 1.0 GB/s, Model 2’s maximum operational intensity (OI = 8 MAC/B) cannot offset data transfer delays, bounding it to the memory roof. Model 3, with a maximum OI of 64 MAC/B, retains sufficient data reuse to remain compute-bound. This confirms that scale-up designs mask poor bandwidth via volumetric retention, yet become inefficient given adequate memory interfaces.

C. Energy-Delay Product Assessment

Table I summarizes the EDP analysis for a standardized 10^6 MAC workload across representative configurations, including a scale-out array of four parallel Model 2 units. Operating at peak throughput, the monolithic Model 3 reduces latency by 7.01 μs compared to a single Model 2, but requires 1790.00 μW to do so — yielding total energy consumption $18\times$ greater. However, when comparing the monolithic approach directly against the distributed $4\times\text{M2}$ array, the scale-out topology proves superior in both speed and power: it completes the workload in just 9.59 μs while consuming 330.40 μW . Model 3 consequently produces an EDP of 17.59×10^{-13} J s, an order of magnitude above the single Model 2’s 1.22×10^{-13} J s, and vastly inferior to the distributed array’s 0.30×10^{-13} J s. Model 1 illustrates the opposing failure mode: undersizing the

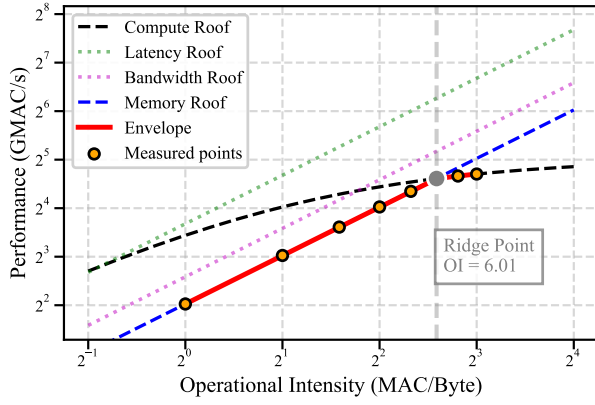


Fig. 3. Extended Roofline Model for Model 2 under the 6 GB/s (10 ns) Memory Constraint.

TABLE I
ENERGY-DELAY PRODUCT COMPARISON FOR A 10^6 MAC WORKLOAD

Arch.	Power (μ W)	Delay (μ s)	Energy (nJ)	EDP (J-s)
M1	54.80	126.90	6.95	8.82×10^{-13}
M2	82.60	38.37	3.17	1.22×10^{-13}
$4 \times M2$	330.40	9.59	3.17	0.30×10^{-13}
M3	1,790.00	31.36	56.11	17.59×10^{-13}

datapath inflates delay to 126.90 μ s, yielding an intermediate EDP of 8.82×10^{-13} J-s that confirms both extremes of the configuration space are suboptimal. Expanding internal register depth degrades the energy-delay balance; distributed scale-out nodes constitute the optimal deployment strategy for energy-constrained edge environments.

Two important qualifications apply to this comparison. First, substituting Model 3 with $4 \times$ Model 2 reduces the effective input window fourfold from 512 to 128, requiring architectural restructuring for applications needing wider receptive fields. Second, the design space was not explored beyond $\text{SIMD}=1024$; a monolithic configuration at $\text{SIMD}=2048$ or higher may represent a more competitive scale-up baseline than Model 3.

V. CONCLUSION

Edge accelerator performance relies on balancing parallel arithmetic execution with available memory bandwidth. The empirical synthesis metrics demonstrate that expanding a monolithic datapath array increases localized data retention, but incurs prohibitive silicon area and thermal costs due to the reliance on discrete registers for parameter storage. Distributing the required matrix operations across multiple dimensionally constrained processing elements optimizes the energy-delay balance. This microarchitectural configuration confirms that a scale-out topology supersedes contiguous scale-up architectures under standard embedded bandwidth limits.

Advancing this evaluation framework involves expanding upon the controlled baselines established. While the current utilization of discrete flip-flop arrays successfully isolates the raw scaling penalties of weight residency, future iterations

replacing these standard registers with integrated SRAM macros will reduce static power dissipation and better reflect the physical storage densities of commercial deployments. Additionally, incorporating Switching Activity Interchange Format waveforms will transition the power estimation from the current statistical baseline to data-dependent, workload-specific dynamic power profiling. Finally, transitioning from a uniform clock target to per-configuration frequency optimization will help establish the absolute performance limits of each distinct geometry.

Although the absolute values depend on the simplifying assumptions adopted in this work, the hardware metrics consistently indicate that scaling internal register depth severely degrades power efficiency. To optimize edge acceleration, designers must restrict localized storage capacities and prioritize distributing computation across dimensionally constrained parallel nodes.

ACKNOWLEDGMENT

A. B. Weber thanks his family and the “Studas” group for their continued support. During manuscript preparation, Gemini 3.1 Pro and NotebookLM were utilized for grammatical refinement and bibliographic verification. The authors assume full responsibility for the final technical content.

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